

Integers, Primes and Roots — Paper D

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to come out exactly.
- The rules for HCF, LCM and signs are not printed on this paper. You are expected to know them.
- You are expected to know the squares up to 15^2 and the cubes up to 10^3 by heart.
- Several questions describe steps you may not have seen before — apply the ideas you already have. If a number will not factorise neatly, ask whether its prime factorisation is telling you something about squares or cubes.
- The mark for each question is shown in square brackets on the right.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – BUILDING AND BREAKING

1. Consider the number 2520.

- (a) Write 2520 as a product of its prime factors, in index form. [2]

- (b) Hence write down the largest square number that is a factor of 2520, and state [2]
what 2520 divided by it comes to.

2. Still using 2520:

- (a) find the smallest positive whole number k such that $2520k$ is a perfect square [3]

(b) hence write down $\sqrt{2520k}$

[1]

SECTION 2 – HCF, LCM AND SIGNS

3. Two whole numbers have an HCF of 15 and an LCM of 90.

(a) Write down the product of the two numbers.

[1]

(b) One of the numbers is 45. Find the other, and verify it using prime factorisations.

[2]

(c) Explain why neither of the two numbers could possibly be 20.

[1]

4. Work out:

(a) $(-2)^5$

[1]

(b) $(-5)^2 - 5^2$

[1]

(c) -3^3

[1]

(d) $-120 \div -6 \times -2$

[1]

5. At 06:00 the temperature on a mountain is 5°C . It then falls by 4°C every hour.

(a) What is the temperature at 11:00?

[1]

(b) At what time does the temperature reach -27°C ?

[2]

(c) Your working for (b) divided a negative by a negative and gave a positive answer. Explain, in terms of the situation, why a positive answer is the right kind of answer.

[1]

SECTION 3 – MARKING SOMEBODY ELSE

6. Two students were each asked to find the LCM of 18 and 24.

[5]

Student P wrote: " $18 \times 24 = 432$, so the LCM is 432."

Student Q wrote: " $18 = 2 \times 3^2$ and $24 = 2^3 \times 3$, so the LCM is $2 \times 3 = 6$."

For each student, explain what they have actually calculated and why it is not the LCM. Then give the correct LCM of 18 and 24.

7. A student has written two answers in his book, and both are wrong. [5]

(i) " $\sqrt{81} = 40.5$, because half of 81 is 40.5."

(ii) " $-4 + -6 = 10$, because two negatives make a positive."

For each one, explain the mistake and give the correct answer. Then state exactly which operations the "two negatives make a positive" rule does apply to.

End of paper. Check every answer has a unit or a form the question actually asked for.